Zhou *et al. J. Wirel. Commun. Netw.* (2026) 2026:93
https://doi.org/10.1186/s13638-026-02637-2

Journal on Wireless Communications and Networking

RESEARCH Open Access

Joint optimization of adaptive semantic compression and resource allocation for semantic-enabled UAV communications

Yuchen Zhou[1], Peng Dai[1*], Bingtao He[1], Mengmeng Ren[1], Mofan Luo[1] and Jian Chen[1]

*Correspondence:
daipeng0916@stu.xidian.edu.cn

[1] The State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China

Abstract

Semantic communication serves as an effective approach for real-time perception in next-generation wireless communication systems. In this paper, we construct a multi-UAV semantic monitoring system with adaptive semantic compression strategy. For video frame transmission in this system, we introduce a novel metric called the Age of Incorrect Semantics (AoIS). Then, we formulate an optimization problem to minimize the total AoIS by jointly optimizing the compression ratio, transmission power, and transmission decisions. Since semantic encoding and decoding cannot be expressed by explicit formulas, conventional convex optimization iterative algorithms cannot solve this problem. To solve this challenging problem, we develop a deep reinforcement learning (DRL) algorithm based on centralized training with decentralized execution (CTDE), where global observations are employed for both critic network training and advantage computation, while local observations are used for policy execution. Simulation results demonstrate that the proposed strategy effectively balances semantic similarity and transmission frequency compared with the baselines.

Keywords: Semantic communication, Semantic compression, Age of information, Resource allocation

1 Introduction

Unmanned aerial vehicles (UAVs) have found increasingly widespread applications in data collection, environmental monitoring, wireless communication, and other fields due to their unique mobility and flexibility [1, 2]. However, constrained by inherent physical limitations such as compact size, limited onboard energy, and highly restricted computing and storage resources [3], UAV communications relying on conventional communication paradigms (i.e., transmitting raw data without semantic compression) face severe challenges, which make it difficult to meet the requirements of real-time and efficient transmission for massive data [4]. Unlike fixed edge servers or static IoT devices, UAVs operate in highly dynamic environments—their mobility causes rapid channel state information (CSI) fluctuations, and they face far stricter energy constraints (limited onboard batteries vs. grid power) [5–7]. Meanwhile, semantic communication, as a new communication paradigm, has been identified as one of the core technologies for the

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Springer

next generation of wireless communication systems [8]. By focusing on the meaning of information, it breaks through the limitations of bit-stream transmission in conventional communications, enabling efficient extraction and reconstruction of data while ensuring the integrity of critical information and greatly saving communication resources [9]. For UAVs, semantic communication's "meaning-first" approach is uniquely valuable: it reduces data volume by compressing redundant information to save energy and bandwidth, and adapts to dynamic CSI–advantages less relevant or beneficial to static edge systems [10].

Previously, many studies have been conducted to efficiently optimize resource allocation in semantic communication. Recent studies [11–13] focused on constructing fundamental frameworks for semantic communication in dynamic environments. Specifically, the authors in [11] proposed a hierarchical federated learning framework to tackle UAV-enabled semantic model training. A cognitive semantic communication system based on knowledge graphs was proposed in [12] to enhance detection performance under low SNR conditions. The authors in [13] proposed a multi-modal semantic communication system for efficient transmission of image and text features. Recent studies have proposed various new metrics in order to improve the performance of semantic communication systems. Recent studies [14–17] focused on communication resource allocation and consider multi-dimensional metrics. The authors in [14] optimized the resource allocation by evaluating the importance of semantic triplets, aiming to maximize total semantic similarity. The authors in [15] optimized the resource allocation by jointly considering channel assignment and the number of semantic symbols transmitted in semantic communication. The authors in [16] proposed a novel quality-of-experience (QoE) metric and applied it to the resource allocation. A dynamic resource allocation method for vehicular networks was proposed in [17], considering metrics like user satisfaction and queue stability to improve transmission efficiency. Existing works focus on the implementation of semantic communication systems, using fixed neural networks for feature extraction without compression. In practice, if no compression is applied to the transmitted data, users may consume excessive communication resource to transmit non-critical data, thus leading to unnecessary resource waste.

To tackle the excessive consumption of communication resources, recent works have explored more efficient semantic information extraction and compression strategies. Recent studies [18–21] focused on achieving semantic compression by designing information importance evaluation methods. The works in [18] and [19] considered self-attention to selectively transmit collaborative information based on the correlation between the key and query or message importance. The authors in [20] adopted collaborative depth estimation to select communication content and each agent evaluates and selects information regions for sharing based on the estimation results. The authors in [21] proposed spatial-channel filtering based on mutual information to extract key features. Some works have further explored the joint optimization of semantic compression and resource allocation [22–24]. The work in [22] adopted the cooperative reconstruction method and a feature compressor to achieve bandwidth optimization. The work in [23] considered the resource allocation problem in the joint source and channel coding (JSCC) system, optimizing users' compression ratios, power allocation, and resource block allocation to maximize the number of users that

successfully receive their requested content with desired quality. The authors in [24] proposed a task-oriented semantic compression method, jointly optimizing resource allocation and compression ratio to maximize task success probability.

While existing works have investigated on the semantic communication resource allocation and semantic compression, these works still exist two major limitations:

Lack of adaptive semantic compression: While existing works have incorporated semantic compression into communication resource optimization, most of these works adopt fixed-compression methods. Specifically, these methods may waste resources by retaining redundant background information in simple scenarios and cause information distortion due to over-compression of features in complex scenarios. Without addressing this problem, the intense competition for limited resources in wireless communication systems will further escalate. We propose an adaptive semantic compression strategy based on real-time perception, which adaptively adjusts the semantic compression ratio according to the importance of semantic features and CSI.

Neglect of information freshness: The above-mentioned works [11–24] on semantic communication systems fail to consider the significance of information freshness. The Age of Information (AoI), a key metric for measuring information freshness, is defined as the time difference between information generation and reception [25]. It has been investigated in semantic communication systems [26–29]. Although these pioneering works demonstrate the potential of integrating AoI with semantic communication, they have neither redefined the AoI metric nor formulated an optimization framework tailored for semantic communication systems. This limitation restricts the joint optimization of semantic information freshness and resource allocation, consequently limiting the performance of semantic communication systems in real-time scenarios. We redefine AoI for semantic communication by proposing the Age of Incorrect Semantics (AoIS) metric, which incorporates semantic similarity into the freshness evaluation framework. Then, we jointly optimize it with resource allocation.

Motivated by the above observations, we propose a UAV-oriented semantic compression strategy, which considers semantic feature importance and freshness. Based on this, we formulate an optimization problem to minimize total AoIS across UAVs by jointly optimizing compression ratio, transmission power, and transmission decision. Then, we propose a deep reinforcement learning (DRL) algorithm based on centralized training with decentralized execution (CTDE) to solve it. The main contributions of this paper can be summarized as follows:

- We propose a novel adaptive semantic compression strategy for our multi-UAV semantic monitoring system. By adjusting the compression ratio in real-time, this method can dynamically scales the volume of transmitted data based on both semantic feature importance and CSI.
- We extend the conventional AoI metric to semantic communication by introducing the AoIS. This metric incorporate semantic similarity into freshness, enabling a more comprehensive evaluation of semantic information value.
- We develop a CTDE-based reinforcement learning algorithm that learns collaborative policies in centralized training and generates actions based on local

observation in decentralized execution. This design eliminates the dependency on global communication while ensuring the effectiveness of local decision.

- Simulation results demonstrate the convergence of the proposed algorithm. The results reveal that the proposed strategy effectively balances transmission frequency with semantic similarity across diverse conditions, highlighting its adaptability in various operational environments.

2 Method

2.1 System model

In this section, we describe the multi-UAV semantic monitoring system, including an adaptive semantic compression module and the AoIS metric. Then, we formulate an optimization problem to minimize the total AoIS, and finally propose a CTDE-based reinforcement learning algorithm to solve it.

2.1.1 System model description

As shown in Fig. 1, we consider a semantic communication monitoring system with multiple UAVs. Each UAV captures a sequence of video frames and transmits them to a ground control station (GCS) via semantic communication. In this transmission process, the semantic encoder equipped on each UAV maps the original video frames to semantic

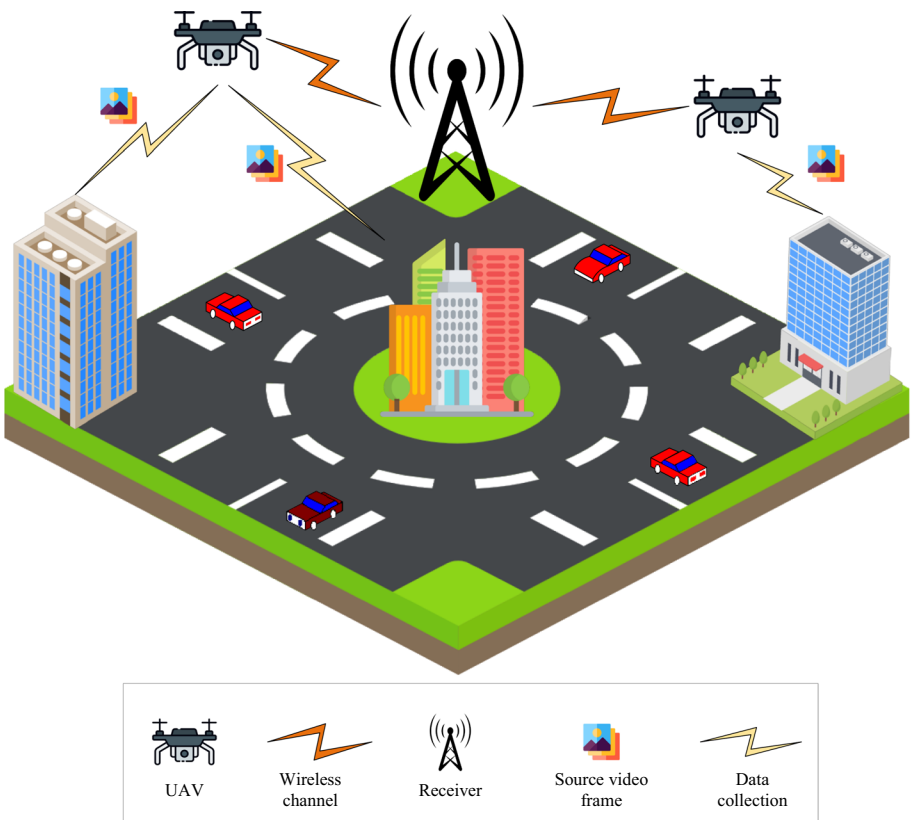


Fig. 1 Multi-UAV semantic monitoring system

features. After the data are transmitted through the wireless channel, the GCS reconstructs the original information via a semantic decoding network.

Figure 2 depicts the system model including K UAVs, each UAV i ($i \in \{1, \ldots, K\}$) synchronously collects video frame data $S_i(t)$ during discrete time slots t ($t \in \{1, \ldots, T\}$). The system uses an end-to-end adaptive semantic communication architecture to achieve dynamic transmission optimization of video data. The transmission decision of each UAV at time slot t is characterized by a binary variable $\alpha_i(t)$:

$$\alpha_i(t) = \begin{cases} 1, & \text{choose to transmit,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

If the current video frame is selected for transmission at time slot t (i.e., $\alpha_i(t) = 1$), UAV i first utilizes its semantic encoder E_s to extract the semantic features of the original video frame. It then performs semantic compression to retain the critical components of these features, reducing the amount of data for wireless transmission. The compressed semantic features are encoded by the channel encoder E_c and transmitted via the wireless channel to the receiver, which reconstructs the original semantic information $\hat{S}_i(t)$ through the channel decoder D_c and semantic decoder D_s. When transmission is not selected at time slot t ($\alpha_i(t) = 0$), the receiver retains the prior reconstructed semantic features $\hat{S}_i(t-1)$. Therefore, the reconstructed semantic feature stored at the receiver for UAV i at time t is given by:

$$\hat{S}_i(t) = \begin{cases} \hat{S}_i(t), & \alpha_i(t) = 1, \\ \hat{S}_i(t-1), & \alpha_i(t) = 0. \end{cases} \tag{2}$$

The semantic similarity $\eta_i(t)$ between the original and reconstructed features is measured using MS-SSIM (Multi-Scale Structural Similarity). For UAV i at time slot t, the similarity metric is given by:

$$\eta_i(t) = \begin{cases} \eta(S_i(t), \hat{S}_i(t)), & \alpha_i(t) = 1, \\ \eta(S_i(t), \hat{S}_i(t-1)), & \alpha_i(t) = 0, \end{cases} \tag{3}$$

where $\eta(\cdot, \cdot)$ denotes the MS-SSIM function. This formulation quantifies the similarity between the current original feature $S_i(t)$ and the reconstructed feature $\hat{S}_i(t)$ when

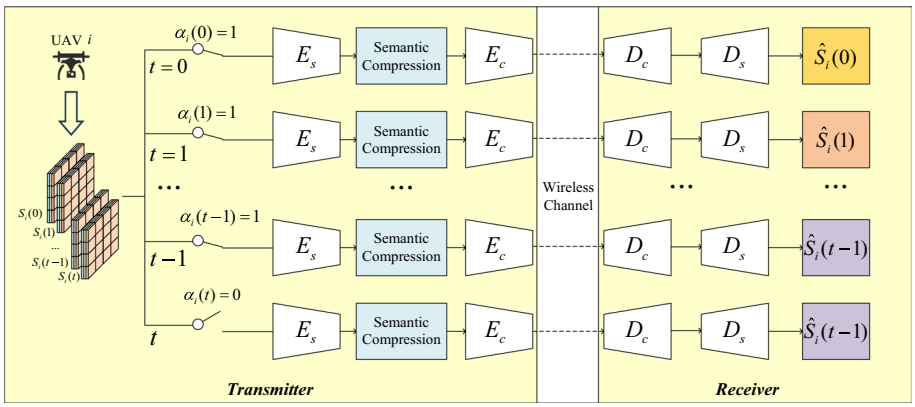


Fig. 2 System model

transmission occurs ($\alpha_i(t) = 1$), or between $S_i(t)$ and the previously stored reconstruction $\hat{S}_i(t-1)$ when transmission is skipped ($\alpha_i(t) = 0$).

2.1.2 Semantic communication with adaptive semantic compression strategy

In the case where UAV i selects to transmit the video frame at the current time slot, the semantic communication system depicted in Fig. 3 is considered. The transmitter comprises three essential modules: the semantic encoder, which extracts semantic features from the source data; the semantic compression module, which reduces the transmitted data volume by compressing features based on their semantic importance; and the channel encoder, which converts the compressed semantic information into transmission-ready symbols. The receiver consists of a channel decoder for decoding received symbols and a semantic decoder for reconstructing the original information.

At the transmitter, a neural network-based semantic encoder extracts semantic features F_i from the original video frame $S_i \in \mathbb{R}^{C\times H\times W}$:

$$F_i = E_s(S_i \mid \sigma), \tag{4}$$

where $E_s(\cdot \mid \sigma)$ denotes the semantic encoder parameterized by σ.

The semantic compression module then performs importance-based selection on the semantic features F_i. Specifically, a multi-layer self-attention network is employed to evaluate the importance of different regions within F_i to precisely identify semantically important areas. This process generates a set of importance weights for each region in F_i through an importance extraction network $P(\cdot)$:

$$C_i = P(F_i), \tag{5}$$

where the importance map $C_i \in [0,1]^{H\times W}$ shares dimension with F_i. The compression ratio ρ is defined as the ratio of nonzero elements in C_i to the total number of elements, mathematically expressed as

$$\rho = \frac{\|C_i\|_0}{H \times W}, \tag{6}$$

where $\|\cdot\|_0$ denotes the zero norm counting nonzero entries.

The semantic features F_i are element-wise multiplied with the generated importance map C_i to yield the compressed semantic features F_i':

$$F_i' = F_i \odot C_i, \tag{7}$$

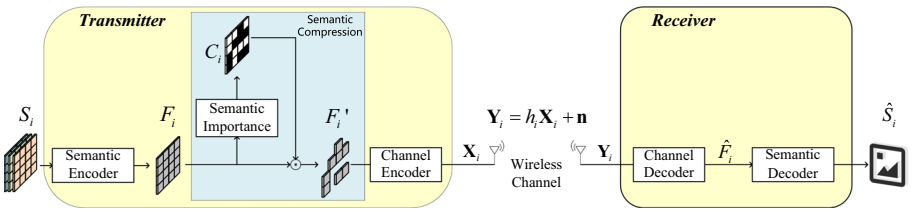


Fig. 3 Semantic communication system with the adaptive semantic compression module

where $F_i' \in \mathbb{R}^{C\times H\times W}$. After element-wise multiplication with the importance map, the semantic features are sparsified, requiring only non-zero elements of F_i' for transmission.

The compressed features F_i' are encoded into a symbol vector $\mathbf{X}_i$ via the channel encoder:

$$\mathbf{X}_i = E_c(F_i' \mid \zeta), \tag{8}$$

where $E_c(\cdot \mid \zeta)$ denotes the channel encoder parameterized by ζ. The symbol vector is transmitted over the wireless channel, and the received signal at the receiver is given by:

$$\mathbf{Y}_i = h_i\mathbf{X}_i + \mathbf{n}, \tag{9}$$

where h_i represents the channel gain and $\mathbf{n}$ denotes additive white Gaussian noise (AWGN).

The received signal $\mathbf{Y}_i$ is decoded by the channel decoder and semantic decoder at the receiver, yielding the reconstructed semantic features $\hat{S}_i(t)$, as given by:

$$\hat{S}_i = D_s\big(D_c(\mathbf{Y}_i \mid \psi) \mid \xi\big), \tag{10}$$

where $D_c(\cdot \mid \psi)$ denotes the channel decoder parameterized by ψ and $D_s(\cdot \mid \xi)$ represents the semantic decoder parameterized by ξ. Finally, the receiver uses the reconstructed semantic features $\hat{S}_i$ to execute subsequent tasks.

2.1.3 Wireless communication

We assume that UAV i flies along a predetermined trajectory at a fixed altitude H_i, and the coordinate of UAV i at time slot t is denoted as $\big(x_i(t), y_i(t), H_i\big)$. The distance between UAV i and the receiver with the coordinate $\big(x_r, y_r, 0\big)$ at time slot t is $z_i(t) = \sqrt{(x_i(t)-x_r)^2 + \big(y_i(t)-y_r\big)^2 + H_i^2}$. The elevation angle between UAV i and the receiver is $\theta_{r,i}(t) = \frac{180}{\pi} \times \arcsin\left(\frac{H_i}{z_i(t)}\right)$.

To capture dynamic Line-of-sight (LoS)/Non-line-of-sight (NLoS) transitions from UAV mobility, we adopt a widely validated LoS probability model tailored for UAV scenarios [30–33]. This model quantifies LoS likelihood as a function of the elevation angle $\theta_{r,i}(t)$, which directly reflects the UAV's relative height and distance to the receiver. Then, the probability of establishing LoS communication link between UAV i and the receiver is

$$P_i^{\mathrm{LoS}}(t) = \frac{1}{1 + A\exp\big(-B(\theta_{r,i}(t) - A)\big)}, \tag{11}$$

where A and B are model parameters. Higher elevation angles from closer or higher UAVs increase LoS probability, while lower angles from distant or descending UAVs lead to more frequent NLoS conditions.

Then, the channel power gain between UAV i and the receiver can be approximately modeled by:

$$h_i(t) = P_i^{\mathrm{LoS}}(t)h_i^{\mathrm{LoS}}(t) + (1 - P_i^{\mathrm{LoS}}(t))h_i^{\mathrm{NLoS}}(t), \tag{12}$$

where

$$h_i^{\mathrm{LoS}}(t) = \frac{\beta_0}{z_i^{\alpha_{\mathrm{L}}}(t)},\ \mathrm{h}_i^{\mathrm{NLoS}}(t) = \frac{\beta_0}{z_i^{\alpha_{\mathrm{N}}}(t)}\varsigma \tag{13}$$

denote, respectively, the channel power gains conditioned on the LoS and NLoS states [34], β_0 denotes the channel power gain at the reference distance $z_0 = 1\,\mathrm{m}$ in the LoS state, α_{L} and α_{N} denote, respectively, the average path loss exponents for the LoS and NLoS states, $\varsigma \sim \mathcal{N}(0, \delta^2)$ is the random variable representing small-scale fading.

According to Shannon's information theory, all transmitted symbols are quantized into discrete bitstreams via binary mapping during physical-layer transmission. Thus, the transmission rate of UAV i is given by:

$$R_i(t) = B_i \log_2\left(1 + \frac{h_i(t)P_i(t)}{N_0 B_i}\right), \tag{14}$$

where B_i denotes the allocated bandwidth, $P_i(t)$ denotes the transmit power, and N_0 denotes the noise power spectral density.

Based on general assumptions in system modeling, the data size of the original semantic features extracted by UAV i at time slot t is denoted as $d_i(t)$, and its semantic compression ratio is defined as $\rho_i(t)$. Consequently, the actual transmitted data size of UAV i can be quantified as $d_i'(t) = d_i(t) \cdot \rho_i(t)$, and the transmission delay is $\tau_i(t) = \frac{d_i'(t)}{R_i(t)}$.

In UAV monitoring scenarios, many tasks are delay-sensitive, necessitating a strict transmission delay constraint $\tau_{\max}$. Therefore, the success transmission probability for UAV i at time slot t is $P_r(\tau_i(t) \le \tau_{\max})$, which is given by:

$$\begin{aligned} P_r(\tau_i(t) \le \tau_{\max}) &= P_r\Big(\frac{d_i'(t)}{R_i(t)} \le \tau_{\max}\Big) \\ &= P_r\Big(\frac{d_i(t)\rho_i(t)}{B_i \log_2\left(1 + \frac{h_i(t)P_i(t)}{N_0 B_i}\right)} \le \tau_{\max}\Big) \\ &= P_r\Big(\frac{N_0 B_i (2^{\frac{d_i(t)\rho_i(t)}{B_i \tau_{\max}}} - 1) z_i^{\alpha_N}(t)}{P_i(t)\beta_0(1 - P_i^{LoS}(t))} - \frac{P_i^{LoS}(t) z_i^{\alpha_N}(t)}{(1 - P_i^{LoS}(t)) z_i^{\alpha_L}(t)} \le \varsigma\Big) \\ &= Q_N\Big(\frac{N_0 B_i (2^{\frac{d_i(t)\rho_i(t)}{B_i \tau_{\max}}} - 1) z_i^{\alpha_N}(t)}{P_i(t)\beta_0(1 - P_i^{LoS}(t))\delta} - \frac{P_i^{LoS}(t) z_i^{\alpha_N}(t)}{(1 - P_i^{LoS}(t)) z_i^{\alpha_L}(t)\delta}\Big) \end{aligned} \tag{15}$$

where the last equality follows from $\varsigma \sim \mathcal{N}(0, \delta^2)$, Q_N is the tail distribution function of the normal distribution. To enhance the transmission success probability, UAV i should increase its transmit power $P_i(t)$ and maintain a relatively low compression ratio $\rho_i(t)$.

2.1.4 Age of incorrect semantics (AoIS) metric

In the classical AoI theory, freshness is only determined by time:

$$\Delta_{\mathrm{AoI}}(t) = t - \varepsilon(t), \tag{16}$$

where $\varepsilon(t)$ denotes the latest update time at the receiver. Conventional AoI has two key limitations: (1) it ignores information correctness, treating incorrect and correct data equally; (2) it does not consider the varying utility of different semantic features. Consequently, conventional AoI cannot effectively reflect information value decay in semantic

communication. To solve the above problems, age of incorrect information (AoII) considers information content:

$$\Delta_{\text{AoII}}(t) = (t - \varepsilon(t))\mathbf{1}\{\hat{S}(t) \neq S(t)\}, \tag{17}$$

where the indicator function $\mathbf{1}\{\cdot\}$ accumulates time for incorrect state estimates. However, it only discriminates between correct and incorrect states, failing to quantify the degree of semantic distortion.

To address the unique needs of semantic communication systems, we propose the age of incorrect semantics (AoIS) metric, defined as:

$$\Delta(t) = \exp\left(b \cdot (t - \varepsilon(t))\right) \cdot \left(1 - \eta(S(t), \hat{S}(t))\right), \tag{18}$$

where $b > 0$ is a positive constant, $\varepsilon(t)$ denotes the last time the receiver receives the information from transmitter. Thus, reducing AoIS requires achieving shorter update intervals and maintaining higher semantic similarity.

2.2 Problem formulation

For each UAV i, the age of incorrect semantics (AoIS) is specialized as:

$$\Delta_i(t) = \exp\left(b \cdot (t - \varepsilon_i(t))\right) \cdot \left(1 - \eta(S_i(t), \hat{S}_i(t))\right), \tag{19}$$

where $\varepsilon_i(t)$ denotes the last time the receiver receives the video frame transmitted by UAV i, defined as:

$$\varepsilon_i(t) = \begin{cases} t, & \alpha_i(t) = 1, \\ \varepsilon_i(t-1), & \alpha_i(t) = 0. \end{cases} \tag{20}$$

In this paper, we aim to jointly optimize the transmission decisions $\alpha_i(t)$, compression ratios $\rho_i(t)$, and transmit powers $P_i(t)$, to minimize the total AoIS across all UAVs and time slots under transmission delay and communication resource constraints. Mathematically, the optimization problem is formulated as:

$$\min_{\alpha_i(t),\, \rho_i(t),\, P_i(t)} \sum_{t=1}^{T} \sum_{i=1}^{K} \Delta_i(t) \tag{21}$$

$$\text{s.t.} \quad \alpha_i(t) = \begin{cases} 1, \text{ choose to transmit}, \\ 0, \text{ otherwise}, \end{cases} \quad \forall i, t, \tag{21a}$$

$$0 \leq \rho_i(t) \leq \alpha_i(t), \quad \forall i, t, \tag{21b}$$

$$P_r(\tau_i(t) \leq \tau_{\max}) \geq P_{r,\min}, \quad \forall i, t, \tag{21c}$$

$$0 \leq P_i(t) \leq P_{\max} \cdot \alpha_i(t), \quad \forall i, t, \tag{21d}$$

$$\sum_{i=1}^{K} P_i(t) \leq P_{\text{total}}, \quad \forall t, \tag{21e}$$

$$\varepsilon_i(t) = \tag{21f}$$

$$\begin{cases} t, & \alpha_i(t) = 1, \\ \varepsilon_i(t-1), & \alpha_i(t) = 0, \end{cases} \quad \forall i, t,$$

$$\eta_i(t) = \begin{cases} \eta(S_i(t), \hat{S}_i(t)), & \alpha_i(t) = 1, \\ \eta(S_i(t), \hat{S}_i(t-1)), & \alpha_i(t) = 0, \end{cases} \quad \forall i, t, \tag{21g}$$

$$\hat{S}_i(t) = \begin{cases} \hat{S}_i(t), & \alpha_i(t) = 1, \\ \hat{S}_i(t-1), & \alpha_i(t) = 0, \end{cases} \quad \forall i, t. \tag{21h}$$

Constraint (21a) defines the binary transmission decision for each UAV. Constraint(21b) ensures no data remain when no transmission is made. Constraint (21c) enforces the minimum transmission success probability. Constraint (21d) limits each UAV's transmit power (zero when not transmitting) with $P_{\max}$ being the maximum transmit power of a single UAV. Constraint (21e) caps the total transmit power across all UAVs, where P_{total} is the maximum total transmit power. Constraints (21f)-(21 h) define the update rules for the last reception time, semantic similarity, and reconstructed features, respectively.

2.3 CTDE-based algorithm design

The multi-UAV optimization problem in this paper is NP-hard due to mixed discrete-continuous variables and high dimensionality, making conventional convex optimization inefficient. To address this, we propose a CTDE-based reinforcement learning algorithm with the centralized critic network (global training) and the decentralized actor network (local action generation), with shared parameters across agents (common objective). Centralized training and centralized execution (CTCE) is avoided due to its reliance on global information, leading to high communication overhead and single-point failures.

Specifically, both the actor and critic networks use a shared 1-layer MLP feature extractor (64 hidden units, ReLU activation, layer normalization) to process inputs, with independent layers for action generation (actor) and value estimation (critic). The actor's output layer adapts to task needs (Categorical for discrete actions, Diag-Gaussian for continuous actions), while the critic uses a linear layer for global Q-value estimation. All layers use orthogonal initialization (gain = 0.01 for outputs) to stabilize training. This shared design ensures consistent feature representation across agents, reduces parameter redundancy, and enhances robustness.

In our algorithm framework, each UAV acts as an independent agent in a distributed manner, forming a partially observable markov decision processes (POMDP) due to unobservable actions of other agents. Core POMDP elements are defined as:

1) Environment state: According to the optimization problem, the environment state at time slot t is:

$$s(t) = \left\{ t, \{\varepsilon_i(t-1)\}_{i=1,\ldots,K}, \{S_i(t)\}_{i=1,\ldots,K}, \left\{\hat{S}_i(t-1)\right\}_{i=1,\ldots,K}, \left\{h_i(t)\right\}_{i=1,\ldots,K} \right\}, \tag{22}$$

where $\varepsilon_i(t-1)$ denotes the last time the receiver receives the information from UAV i, $S_i(t)$ denotes the original video frame collected by UAV i, $\hat{S}_i(t-1)$ denotes the last reconstructed semantic feature stored at the receiver for UAV i, and $h_i(t)$ denotes the channel power gain between UAV i and the receiver.

2) Observation: The observation of UAV i at time slot t is:

$$o_i(t) = \left\{ t, \varepsilon_i(t-1), S_i(t), \hat{S}_i(t-1), h_i(t) \right\}. \tag{23}$$

3) Action: Based on observations at each time slot, each UAV needs to select transmission decision $\alpha_i(t)$, compression ratio $\rho_i(t)$, and communication power $P_i(t)$. The action of UAV i at time slot t is:

$$a_i(t) = \{\alpha_i(t), \rho_i(t), P_i(t)\}. \tag{24}$$

4) Reward: Considering the integrity of the system optimization goal, all UAVs need to collaborate under resource and transmission decision constraints to maximize the total age of incorrect semantics (AoIS). Thus, the framework designs the same immediate reward function for each agent, given by (25). Here, $R_i(t)$ denotes the reward of UAV i, as given by (26):

$$R(t) = \sum_{i=1}^{K} R_i(t) - R_{\text{penalty}}(t), \tag{25}$$

$$R_i(t) = -\Delta_i(t), \tag{26}$$

where $R_{\text{penalty}}(t)$ is the penalty for violating the constraints.

In the centralized training stage, the critic network acts as a central coordinator to compute the centralized action-value function $Q(s, a_1, a_2, \cdots, a_K|\phi)$ based on global state information containing all UAVs' actions and observations. The Q-function evaluates the actions generated by actor networks from a global perspective and optimizes the multi-agent collaborative decision-making process by guiding policy gradient updates. The critic network then iteratively updates its network parameters θ^c by minimizing the loss function constructed from the temporal difference (TD) error:

$$\mathcal{L}(\theta^c) = \mathbb{E}[Q(s, a_1, a_2, \cdots, a_K|\phi) - y]^2, \tag{27}$$

where $y = R + \gamma Q'(s', a'_1, a'_2, \cdots, a'_K|\phi')$ is the TD target, s is the global environment state, and the target network parameters ϕ are periodically updated by the critic parameters to stabilize training. Here, s' and ϕ' denote the updated state and network parameters. The actor networks are optimized using a policy gradient algorithm with the loss function defined as:

$$\mathcal{L}(\theta_i) = -\mathbb{E}_{\tau\sim\pi_{\theta_i}}\left[\text{clip}\left(\frac{\pi_{\theta_i}(a_i|o_i)}{\pi_{\theta_i^{\text{old}}}(a_i|o_i)}, 1-\epsilon, 1+\epsilon\right)\cdot\hat{A}_i(s,\mathbf{a})\right] - c_1\cdot\mathcal{H}(\pi_{\theta_i}(\cdot|o_i)), \tag{28}$$

where θ_i is the parameter of the actor network for the UAV i, clip$(\cdot)$ is the clipping function, $\pi_{\theta_i}(a_i|o_i)$ is the current policy, $\pi_{\theta_i^{\text{old}}}(a_i|o_i)$ is the old policy before update, ϵ is the clipping parameter limiting the policy update step size,

$$\hat{A}_i(s,\mathbf{a}) = Q(s,a_1,a_2,\cdots,a_K|\phi) - V_\phi(s) \tag{29}$$

is the advantage function computed based on the centralized critic network, $V_\phi(s)$ is the state-value function, $\mathcal{H}(\pi_{\theta_i}(\cdot|o_i))$ is the policy entropy, and c_1 is the policy entropy weight. The actor networks update their parameters according to the centralized Q-function from the critic and their own observations, then output actions. Specifically, each actor network updates its parameters θ_i in the direction of $\nabla_{\theta_i}J(\theta_i)$ as:

$$\nabla_{\theta_i}J(\theta_i) \approx \mathbb{E}\left[\nabla_a Q(s,a_1,a_2,\cdots,a_K|\phi)\nabla_{\theta_i}\pi(s|\theta_i)\right]. \tag{30}$$

In the decentralized execution stage, each UAV generates action decisions only via its independent actor network π_{θ_i} based on local observations o_i:

$$a_i = \pi_{\theta_i}(a_i|o_i). \tag{31}$$

This process involves only forward propagation, significantly reducing decision latency and computational cost compared to the training stage. Although lacking other UAVs' information, trained actor networks still generate near global optimal actions. The detailed process is described in Algorithm 1.

Algorithm 1 CTDE-based Algorithm

Initialize actor network π_{θ_i} and critic network V_ϕ
Initialize a memory buffer D
for episode = 1, 2, …, L **do**
  Initial state $\mathbf{s}$ and buffer D
  for step = 1, 2, …, T **do**
    UAVs choose action $\mathbf{a}_t$ according to $\pi_{\theta_i^{\text{old}}}(a_i|o_i)$ of each UAV i
    UAVs execute action $\mathbf{a}_t$
    UAVs obtain next state $\mathbf{s}_{t+1}$ and reward r_t
    Store ($\mathbf{s}_t$,$\mathbf{a}_t$,r_t,$\mathbf{s}_{t+1}$) into buffer D
    Compute the advantage $A(\mathbf{s}_t,\mathbf{a}_t)$ according to (29)
  end for
  for j = 1, 2, …, J **do**
    Shuffle D and split into minibatches of size B
    for each minibatch $B \subseteq D$ **do**
      Compute the loss according to (27) and (28)
      Apply gradient ascent on θ using by Adam
      Apply gradient descent on ϕ using by Adam
    end for
  end for
  Update network parameters for each UAV
  Empty D
end for

3 Results and discussion

We evaluate the proposed method on the open-source OpenVideo dataset, which contains 672 h of video clips covering scenes such as nature, cities, and people. We extract some videos from it and cropped each frame to a dimension of $128 \times 128 \times 3$. Detailed parameter settings are presented in Table 1. All simulations and experiments are conducted on a computer running Ubuntu 20.04 + CUDA 12.0, equipped with an NVIDIA RTX3090 GPU and using the PyTorch deep learning framework.

The performance of the proposed method is compared with the following method: 1) Heterogeneous agent proximal policy optimization with Beta distribution (Beta-HAPPO) [35], where the CTCE is employed; 2) Standard semantic communication method (SSC) [36], where raw semantic features are transmitted directly without any compression; 3) Fixed compression ratio method (FCR) [24], where a preset fixed compression ratio ρ is applied to semantic features.

To demonstrate the optimality of the proposed algorithm, Fig. 4 compares its convergence performance with two baselines: Beta-HAPPO, and the exhaustive search based algorithm that enumerates all possible combinations to derive the optimal solution. The proposed algorithm rapidly achieves convergence of reward with increasing iterations, demonstrating its effectiveness in policy search. The semi-transparent shadow band (min-max range across 5 runs) verifies its training stability. In contrast, Beta-HAPPO exhibits slower convergence due to its centralized optimization. Although it is closer to the optimal value in later stages, this method demonstrates significantly greater volatility and its resource consumption is significantly higher than the proposed algorithm.

Figure 5 shows the total age of incorrect semantics (AoIS) versus maximum total transmit power under different numbers of UAVs ($K = 3, 4, 5$). The total AoIS exhibits two stages: it decreases with increasing power below a threshold (e.g., 45 mW for $K = 3$, 60 mW for $K = 4$) and remains unchanged above the threshold. The reasons for this observation can be explained as follows. Higher power below the threshold enhances power allocation of each UAV, improving its transmission frequency and semantic similarity. However, when the maximum total transmit power is higher than a certain level, each UAV's power reaches the maximal limit. Therefore, the higher total power doesn't increase individual UAV power, limiting further AoIS reduction.

Table 1 Simulation parameters

Parameter	Value
The number of time slots, T	100
The number of UAVs, K	3–5
Maximum transmit power of each UAV i, $P_{\max}$	15 mW
Bandwidth of each UAV i, B_i	1 MHz
Noise power spectral density, N_0	− 174 dBm/Hz
Epoch	15
Batchsize	32
Optimizer	Adam
Learning rate	$\times 10^{-4}$
Clipping parameter, ϵ	0.2

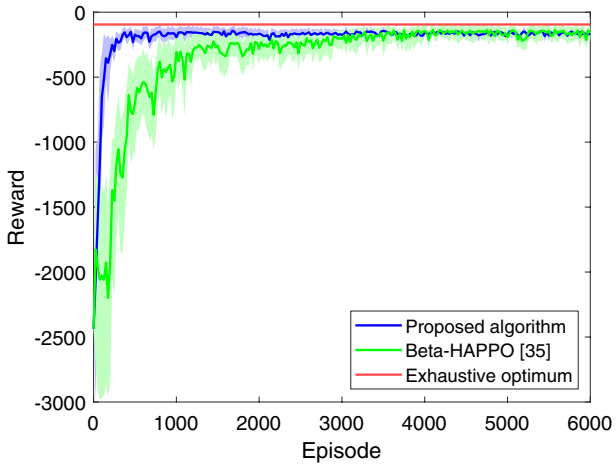


Fig. 4 Convergence performance of the proposed algorithm and Beta-HAPPO

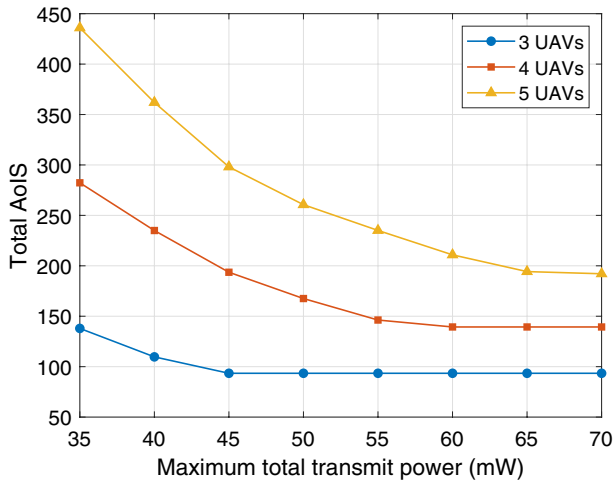


Fig. 5 Total AoIS versus maximum total transmit power under different numbers of UAVs

Figure 6 compares the total AoIS of the proposed strategy, SSC, and FCR with different fixed ρ ($\rho = 0.8, 0.7, 0.5$). It can be observed that SSC has a significantly higher total AoIS than others in the low-power region; FCR with $\rho = 0.8$ outperforms SSC, $\rho = 0.7$ balances frequency and semantic similarity, $\rho = 0.5$ performs well in the low-power region but deteriorates in the high-power region due to excessive compression; our strategy maintains optimal performance across all power ranges. The reasons for these observations are as follows: SSC's poor performance stems from most UAVs failing to meet transmission delay constraints; FCR variants' behavior reflects trade-offs between compression ratio, transmission frequency, and semantic similarity. By adaptively adjusting the compression ratio, our strategy reduces AoIS by 50% to 90% compared to SSC and FCR methods, thereby maintaining optimal maintaining optimal performance across all power ranges.

Figure 7 shows the comparison of the average MS-SSIM of the proposed strategy, SSC, and FCR. It can be observed that SSC performs poorly in the low-power region but

improves in the high-power region; FCR with $\rho = 0.8$ slightly outperforms SSC in the low-power region; FCR with $\rho = 0.7$ surpasses other fixed methods; and FCR with $\rho = 0.5$ saturates in the high-power region. The reasons for these observations are as follows: SSC's low-power limitation stems from insufficient feature updates, FCR variants' performance differences arise from their data retention trade-offs. Conversely, our strategy leverages adaptive compression ratio adjustment, which improves MS-SSIM by 5% to 25% over SSC and FCR methods.

Figure 8 shows the comparison of the average success transmission probability of the proposed strategy, SSC, and FCR. It can be observed that SSC and FCR with fixed ρ show reduced success probability at higher compression ratios. The reasons for this observation are as follows: Higher compression ratios in fixed strategies increase data volume, violating delay constraints and lowering success probability. Consequently, our strategy increases the transmission success probability by 10% to 300% over fixed-compression methods, thereby guaranteeing consistently high transmission rates across nearly all power regions.

4 Conclusion

In this paper, we have developed a multi-UAV semantic monitoring system with adaptive semantic compression. We have introduced a novel metric called age of incorrect semantics (AoIS), which integrates semantic similarity and information freshness. Then, we have formulated an optimization problem to minimize the total AoIS under power and delay constraints, by jointly considering the compression ratio, transmission power, and transmission decisions. Subsequently, we propose a CTDE-based reinforcement learning algorithm to solve this optimization problem, where centralized training discovers the optimal policy under global observations, and decentralized execution generates actions based on local observations. Simulation results have demonstrated the convergence of the proposed method, which adaptively reduces the compression ratio to ensure transmission frequency under low power conditions and increases the compression ratio to improve semantic similarity under high power conditions. These results

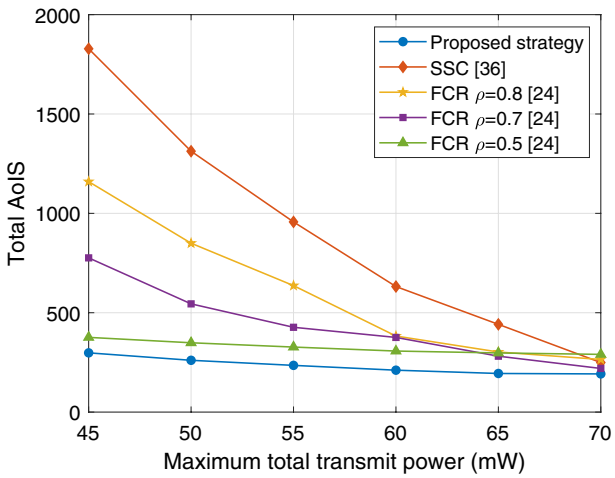


Fig. 6 Total AoIS versus maximum total transmit power under different compression ratios

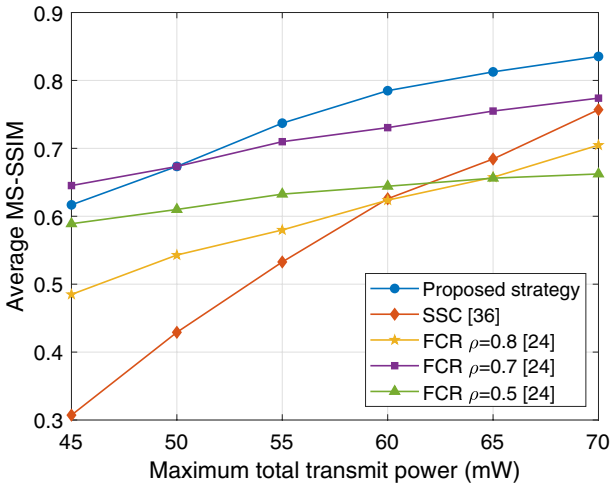


Fig. 7 Average MS-SSIM versus maximum total transmit power under different compression ratios

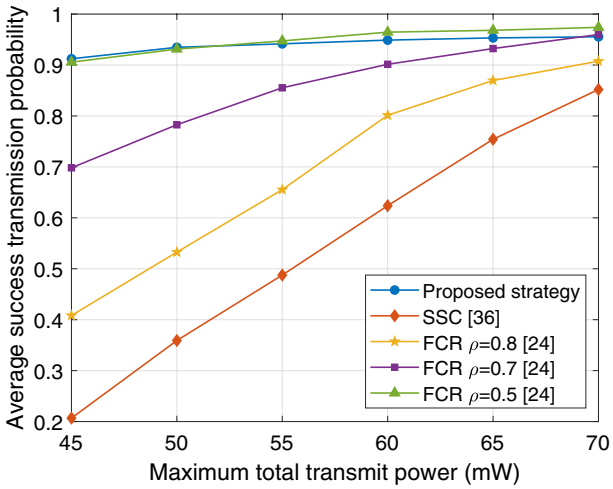


Fig. 8 Average success transmission probability versus maximum total transmit power under different compression ratios

highlight the substantial potential of adaptive semantic compression in improving the efficiency of future wireless communication systems.

Abbreviations

UAV	Unmanned aerial vehicle
QoE	Quality-of-experience
JSCC	Joint source and channel coding
CSI	Channel state information
AoI	Age of Information
AoIS	Age of Incorrect Semantics
DRL	Deep reinforcement learning
CTDE	Centralized training with decentralized execution
GCS	Ground control station
AWGN	Additive white Gaussian noise
LoS	Line-of-sight
NLoS	Non-line-of-sight
AoII	Age of incorrect information

CTCE	Centralized training and centralized execution
POMDP	Partially observable markov decision processes
TD	Temporal difference
Beta-HAPPO	Heterogeneous agent proximal policy optimization with Beta distribution
SSC	Standard semantic communication method
FCR	Fixed compression ratio method

Author contributions
All authors have contributed equally. All authors read and approved the final manuscript.

Funding
This work was supported in part by the National Natural Science Foundation of China under Grant 62371354, Grant 62371367, and Grant 62201421, and in part by the Key Industrial Innovation Chain Project in Industrial Domain under Grant 2023-ZDLGY-50.

Data availability
Not applicable.

Declarations

Competing interests
The authors declare that they have no conflict of interest.

Received: 28 August 2025 Accepted: 13 May 2026
Published online: 29 May 2026

References

1. J. Liu, X. Zhao, P. Qin, S. Geng, Z. Chen, H. Zhou, Learning-based multi-UAV assisted data acquisition and computation for information freshness in WPT enabled space-air-ground PIOT. IEEE Trans. Netw. Sci. Eng. **11**(1), 48–63 (2024). https://doi.org/10.1109/TNSE.2023.3288990
2. L. Zhang, F. Gao, F. Deng, L. Xi, J. Chen, Distributed estimation of a layered architecture for collaborative air–ground target geolocation in outdoor environments. IEEE Trans. Industr. Electron. **70**(3), 2822–2832 (2023). https://doi.org/10.1109/TIE.2022.3165245
3. S. Lu, F. Liu, Y. Li, K. Zhang, H. Huang, J. Zou, X. Li, Y. Dong, F. Dong, J. Zhu, Y. Xiong, W. Yuan, Y. Cui, L. Hanzo, Integrated sensing and communications: Recent advances and ten open challenges. IEEE Internet Things J. **11**(11), 19094–19120 (2024). https://doi.org/10.1109/JIOT.2024.3361173
4. C. She, C. Sun, Z. Gu, Y. Li, C. Yang, H.V. Poor, B. Vucetic, A tutorial on ultrareliable and low-latency communications in 6G: integrating domain knowledge into deep learning. Proc. IEEE **109**(3), 204–246 (2021). https://doi.org/10.1109/JPROC.2021.3053601
5. C. Tan, D. Cai, F. Fang, Z. Ding, P. Fan, Federated unfolding learning for csi feedback in distributed edge networks. IEEE Trans. Commun. **73**(1), 410–424 (2025). https://doi.org/10.1109/TCOMM.2024.3429170
6. C. Tan, D. Cai, Y. Xu, Z. Ding, P. Fan, Threshold-enhanced hierarchical spatial non-stationary channel estimation for uplink massive MIMO systems. IEEE Trans. Wireless Commun. **23**(5), 4830–4844 (2024). https://doi.org/10.1109/TWC.2023.3322984
7. D. Cai, P. Fan, Q. Zou, Y. Xu, Z. Ding, Z. Liu, Active device detection and performance analysis of massive non-orthogonal transmissions in cellular Internet of Things. Sci. China Inf. Sci. **65**, 182301 (2022). https://doi.org/10.1007/s11432-021-3328-y
8. S. Yao, K. Niu, S. Wang, J. Dai, Semantic coding for text transmission: an iterative design. IEEE Trans. Cogn. Commun. Netw. **8**(4), 1594–1603 (2022). https://doi.org/10.1109/TCCN.2022.3192407
9. J. Xu, B. Ai, W. Chen, A. Yang, P. Sun, M. Rodrigues, Wireless image transmission using deep source channel coding with attention modules. IEEE Trans. Circuits Syst. Video Technol. **32**(4), 2315–2328 (2022). https://doi.org/10.1109/TCSVT.2021.3082521
10. G. Chen, D. Cai, Z. Khan, J. Ahmad, W. Boulila, Nash equilibrium between consumer electronic devices and dos attacker for distributed iot-enabled rse systems. IEEE Trans. Consum. Electron. (2025). https://doi.org/10.1109/TCE.2025.3602047
11. J. Xu, H. Yao, R. Zhang, T. Mai, S. Huang, S. Guo, Federated learning powered semantic communication for UAV swarm cooperation. IEEE Wirel. Commun. **31**(4), 140–146 (2024). https://doi.org/10.1109/MWC.005.2300147
12. X. Song, L. Yuan, Z. Qu, F. Zhou, Q. Wu, T.Q.S. Quek, R.Q. Hu , Knowledge graph driven uav cognitive semantic communication systems for efficient object detection. In: ICC 2024 - IEEE International Conference on Communications, pp. 1685–1690 (2024). https://doi.org/10.1109/ICC51166.2024.10622874
13. H. Hu, X. Zhu, F. Zhou, W. Wu, R.Q. Hu, Semantic-oriented resource allocation for multi-modal uav semantic communication networks. In: GLOBECOM 2023 - 2023 IEEE Global Communications Conference, pp. 7213–7218 (2023). https://doi.org/10.1109/GLOBECOM54140.2023.10437885
14. Y. Wang, M. Chen, T. Luo, W. Saad, D. Niyato, H.V. Poor, S. Cui, Performance optimization for semantic communications: an attention-based reinforcement learning approach. IEEE J. Sel. Areas Commun. **40**(9), 2598–2613 (2022). https://doi.org/10.1109/JSAC.2022.3191112
15. H. Xie, Z. Qin, G.Y. Li, Task-oriented multi-user semantic communications for VQA. IEEE Wirel. Commun. Lett. **11**(3), 553–557 (2022). https://doi.org/10.1109/LWC.2021.3136045

16. L. Yan, Z. Qin, R. Zhang, Y. Li, G. Ye Li , Qoe-aware resource allocation for semantic communication networks. In: GLOBECOM 2022 - 2022 IEEE Global Communications Conference, pp. 3272–3277 (2022). https://doi.org/10.1109/GLOBECOM48099.2022.10001594
17. J. Su, Z. Liu, Y.-A. Xie, K. Ma, H. Du, J. Kang, D. Niyato, Semantic communication-based dynamic resource allocation in d2d vehicular networks. IEEE Trans. Veh. Technol. **72**(8), 10784–10796 (2023). https://doi.org/10.1109/TVT.2023.3257770
18. Y-C. Liu, J. Tian, N. Glaser, Z. Kira, When2com: Multi-agent perception via communication graph grouping. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4105–4114 (2020). https://doi.org/10.1109/CVPR42600.2020.00416
19. Q. Yuan, X. Fu, Z. Li, G. Luo, J. Li, F. Yang, Graphcomm: efficient graph convolutional communication for multiagent cooperation. IEEE Internet Things J. **8**(22), 16359–16369 (2021). https://doi.org/10.1109/JIOT.2021.3097947
20. Y. Hu, Y. Lu, R. Xu, W. Xie, S. Chen, Y. Wang, Collaboration helps camera overtake lidar in 3d detection. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9243–9252 (2023). https://doi.org/10.1109/CVPR52729.2023.00892
21. D. Yang, K. Yang, Y. Wang, J. Liu, Z. Xu, R. Yin, P. Zhai, L. Zhang, How2comm: communication-efficient and collaboration-pragmatic multi-agent perception. Adv. Neural Inf. Process. Syst. **36**, 25151–25164 (2023)
22. B. Wang, L. Zhang, Z. Wang, Y. Zhao, T. Zhou, Core: Cooperative reconstruction for multi-agent perception. In: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 8676–8686 (2023). https://doi.org/10.1109/ICCV51070.2023.00800
23. K. Chi, Q. Yang, Z. Yang, Y. Duan, Z. Zhang, Resource allocation for capacity optimization in joint source-channel coding systems. In: ICC 2023 - IEEE International Conference on Communications, pp. 2099–2104 (2023). https://doi.org/10.1109/ICC45041.2023.10279466
24. C. Liu, C. Guo, Y. Yang, N. Jiang, Adaptable semantic compression and resource allocation for task-oriented communications. IEEE Trans. Cognit. Commun. Network. **10**(3), 769–782 (2024). https://doi.org/10.1109/TCCN.2023.3346481
25. S. Kaul, R. Yates, M. Gruteser, Real-time status: How often should one update? In: 2012 Proceedings IEEE INFOCOM, pp. 2731–2735 (2012). https://doi.org/10.1109/INFCOM.2012.6195689
26. J. Chen, J. Wang, C. Jiang, J. Wang, Age of incorrect information in semantic communications for noma aided xr applications. IEEE J. Sel. Top. Signal Process. **17**(5), 1093–1105 (2023). https://doi.org/10.1109/JSTSP.2023.3282836
27. Z.Q. Liew, M. Xu, W.Y.B. Lim, Z. Xiong, D. Niyato, D.I. Kim, Mechanism design for semantic communication in uav-assisted metaverse: A combinatorial auction approach. IEEE Trans. Veh. Technol. **73**(2), 2236–2251 (2024). https://doi.org/10.1109/TVT.2023.3317069
28. W. Wu, Y. Yang, Y. Deng, A.H. Aghvami, Goal-oriented semantic communications for robotic waypoint transmission: The value and age of information approach. IEEE Trans. Wireless Commun. **23**(12), 18903–18915 (2024). https://doi.org/10.1109/TWC.2024.3424493
29. A. Zakeri, M. Moltafet, M. Codreanu, Semantic-aware sampling and transmission in real-time tracking systems: A pomdp approach. IEEE Transactions on Communications, 1–1 (2024) https://doi.org/10.1109/TCOMM.2024.3511935
30. G. Zhang, Q. Wu, M. Cui, R. Zhang, Securing uav communications via joint trajectory and power control. IEEE Trans. Wireless Commun. **18**(2), 1376–1389 (2019). https://doi.org/10.1109/TWC.2019.2892461
31. Q. Wu, Y. Zeng, R. Zhang, Joint trajectory and communication design for multi-uav enabled wireless networks. IEEE Trans. Wireless Commun. **17**(3), 2109–2121 (2018). https://doi.org/10.1109/TWC.2017.2789293
32. Y. Liu, J. Xie, C. Xing, S. Xie, X. Luo, Self-organization of uav networks for maximizing minimum throughput of ground users. IEEE Trans. Veh. Technol. **73**(8), 11743–11755 (2024). https://doi.org/10.1109/TVT.2024.3369020
33. C. You, X. Peng, R. Zhang, 3d trajectory design for uav-enabled data harvesting in probabilistic los channel. 2019 IEEE Global Communications Conference (GLOBECOM), 1–6 (2019) https://doi.org/10.1109/GLOBECOM38437.2019.9014325
34. C. You, R. Zhang, Hybrid offline-online design for uav-enabled data harvesting in probabilistic los channels. IEEE Trans. Wireless Commun. **19**(6), 3753–3768 (2020). https://doi.org/10.1109/TWC.2020.2978073
35. W. Liu, Y. Fu, Y. Guo, F.L. Wang, W. Sun, Y. Zhang, Two-timescale synchronization and migration for digital twin networks: a multi-agent deep reinforcement learning approach. IEEE Trans. Wireless Commun. **23**(11), 17294–17309 (2024). https://doi.org/10.1109/TWC.2024.3452689
36. C.-H. Lee, J.-W. Lin, P.-H. Chen, Y.-C. Chang, Deep learning-constructed joint transmission-recognition for internet of things. IEEE Access **7**, 76547–76561 (2019). https://doi.org/10.1109/ACCESS.2019.2920929

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.